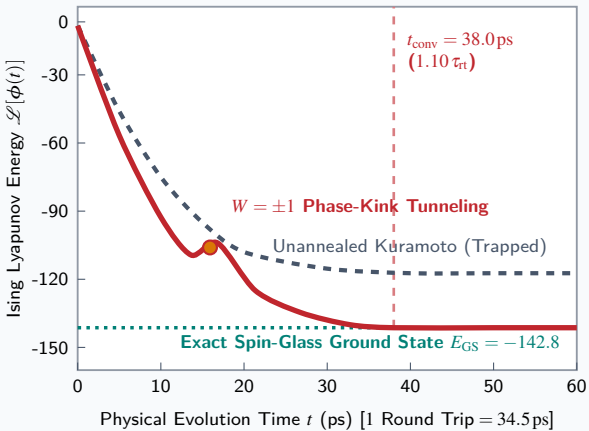
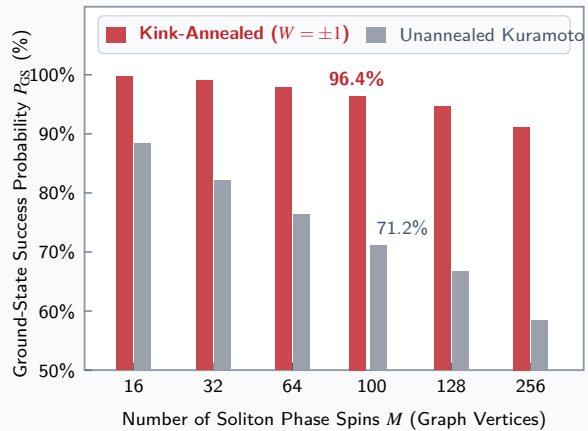

(a) Lyapunov Convergence ($M = 100$ Spins, $t_{\text{conv}} = 38\,\text{ps}$)
$t_{\text{conv}} = 38.0\,\text{ps}$
$(1.10\,\tau_{\text{rt}})$
$W = \pm 1$ Phase-Kink Tunneling
Unannealed Kuramoto (Trapped)
Exact Spin-Glass Ground State $E_{\text{GS}} = -142.8$
Ising Lyapunov Energy $\mathscr{L}[\phi(t)]$
Physical Evolution Time t (ps) [1 Round Trip = 34.5 ps]
0
-30
-60
-90
-120
-150
0
10
20
30
40
50
60


(b) MAX-CUT Ground-State Probability Across Graph Sizes
Kink-Annealed ($W = \pm 1$)
Unannealed Kuramoto
96.4%
71.2%
Ground-State Success Probability P_{GS} (%)
100%
90%
80%
70%
60%
50%
16
32
64
100
128
256
Number of Soliton Phase Spins M (Graph Vertices)